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1. Introduction

The Electronic Dice is a modern, digital alternative to the traditional physical dice used in games. It is designed using basic electronic components such as an Arduino UNO, push button, LEDs or a 7-segment display, and a buzzer. This project simulates the random behavior of a physical dice roll using a microcontroller to generate numbers from 1 to 6. When the user presses a button, the dice "rolls" and a random number is displayed, accompanied by a buzzer sound. This electronic system offers improved durability and eliminates the possibility of cheating. It is especially useful for educational purposes, allowing students to understand key concepts in electronics, coding, and microcontroller-based control. The project is simple, engaging, and practical, making it ideal for learning embedded system basics.

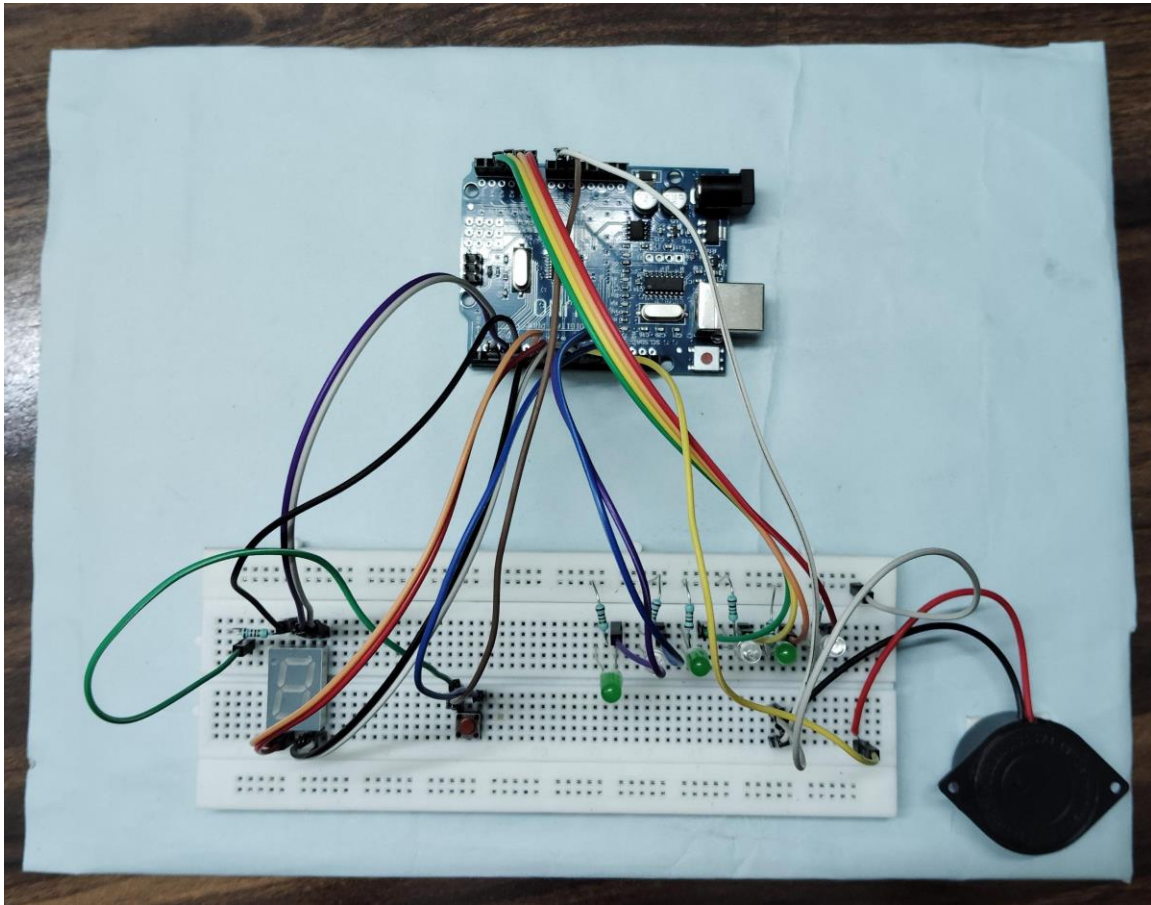
2. Objective

The primary objective of the Electronic Dice project is to create a digital system that can simulate the function of a traditional dice using electronic components. This project aims to generate random numbers between 1 and 6 using an Arduino UNO and display them either through LEDs & a 7-segment display. The project also aims to enhance practical knowledge of hardware interfacing, circuit design, and the use of input/output devices like buttons and buzzers. Another goal is to provide an engaging and fun way to learn embedded systems and Arduino programming. It also encourages creativity and hands-on learning through project-based implementation.

3. Components Used

Component	Quantity	Purpose
Arduino UNO	1	Microcontroller unit
Breadboard	1	Prototyping platform
7-Segment Display	1	Displays numbers 1–6
LEDs	6	Visual representation of dice number
Push Button	1	User input
Buzzer	1	Sound output (dice roll effect)
Jumper Wires	~20	Electrical connections
Resistors (220Ω)	7	Protect LEDs and 7-segment

4. Circuit Diagram



5. Working Principle

The Electronic Dice operates on the basic principle of random number generation using a microcontroller. When the push button is pressed, the Arduino UNO is triggered to generate a random number between 1 and 6, simulating the throw of a dice. This number is displayed using a 7-segment display or LEDs arranged in a dice-like pattern. The randomness is achieved through Arduino's built-in `random()` function, often seeded with an analog input to improve variability. The microcontroller quickly cycles through numbers while the

button is pressed, and the final number is latched upon release. A buzzer is activated for a short sound to mimic the physical interaction of rolling a dice. This principle ensures fairness, randomness, and a user-friendly digital alternative to the traditional dice.

6. Arduino Code

```
const int A = 7;

const int B = 6;

const int C = 4;

const int D = 3;

const int E = 2;

const int F = 8;

const int G = 9;

const int buzzer = 10; // Buzzer connected to pin 10

const int leds[6] = {A0, A1, A2, A3, 11, 13};

int random_int = 0;

void setup() {

  pinMode(A, OUTPUT);

  pinMode(B, OUTPUT);

  pinMode(C, OUTPUT);

  pinMode(D, OUTPUT);

  pinMode(E, OUTPUT);

  pinMode(F, OUTPUT);
```

```

pinMode(G, OUTPUT);

pinMode(12, INPUT_PULLUP);

pinMode(buzzer, OUTPUT); // Buzzer output pin

for (int i = 0; i < 6; i++) {
    pinMode(leds[i], OUTPUT);
}
}

void lightLEDs(int count) {
for(int i = 0; i < 6; i++) {
    if (i < count) {
        digitalWrite(leds[i], HIGH);
    } else {
        digitalWrite(leds[i], LOW);
    }
}
}

void loop() {
    int pushed = digitalRead(12);
    if (pushed == LOW) {
        //digitalWrite(buzzer, HIGH); // Buzzer ON
        //delay(100);
        //digitalWrite(buzzer, LOW); // Buzzer OFF
    }
}

```

```

for (int i = 0; i < 8; i++) {
  int freq = random(300, 1000); // random pitch
  tone(buzzer, freq, 50);    // play tone for 50 ms
  delay(60);                // small pause
}

noTone(buzzer); // stop sound

  random_int = random(1, 7); // Random number from 1 to 6
  lightLEDs(random_int);
  one(); delay(20);
  two(); delay(20);
  three(); delay(20);
  four(); delay(20);
  five(); delay(20);
  six(); delay(20);
}

else {
  switch(random_int) {
    case 1: one(); break;
    case 2: two(); break;
    case 3: three(); break;
    case 4: four(); break;
    case 5: five(); break;

```

```
        case 6: six(); break;
    }
    lightLEDs(random_int); // keep LEDs lit
    delay(200);
}
}
void one(){
    digitalWrite(A, LOW);
    digitalWrite(B, HIGH);
    digitalWrite(C, HIGH);
    digitalWrite(D, LOW);
    digitalWrite(E, LOW);
    digitalWrite(F, LOW);
    digitalWrite(G, LOW);
}
void two(){
    digitalWrite(A, HIGH);
    digitalWrite(B, HIGH);
    digitalWrite(C, LOW);
    digitalWrite(D, HIGH);
    digitalWrite(E, HIGH);
    digitalWrite(F, LOW);
```

```
    digitalWrite(G, HIGH);  
}  
  
void three(){  
    digitalWrite(A, HIGH);  
    digitalWrite(B, HIGH);  
    digitalWrite(C, HIGH);  
    digitalWrite(D, HIGH);  
    digitalWrite(E, LOW);  
    digitalWrite(F, LOW);  
    digitalWrite(G, HIGH);  
}  
  
void four(){  
    digitalWrite(A, LOW);  
    digitalWrite(B, HIGH);  
    digitalWrite(C, HIGH);  
    digitalWrite(D, LOW);  
    digitalWrite(E, LOW);  
    digitalWrite(F, HIGH);  
    digitalWrite(G, HIGH);  
}  
  
void five(){  
    digitalWrite(A, HIGH);
```

```
digitalWrite(B, LOW);  
digitalWrite(C, HIGH);  
digitalWrite(D, HIGH);  
digitalWrite(E, LOW);  
digitalWrite(F, HIGH);  
digitalWrite(G, HIGH);  
}  
  
void six(){  
    digitalWrite(A, HIGH);  
    digitalWrite(B, LOW);  
    digitalWrite(C, HIGH);  
    digitalWrite(D, HIGH);  
    digitalWrite(E, HIGH);  
    digitalWrite(F, HIGH);  
    digitalWrite(G, HIGH);  
}
```

7. Output / Result

The Electronic Dice successfully simulates the rolling of a standard dice by generating a random number between 1 and 6 whenever the push button is pressed. The output is visually represented either by glowing LEDs corresponding to the number or via a 7-segment display showing the digit. A buzzer sound is also produced at the moment of pressing the switch,

enhancing the interactive feel of the dice roll. The device reliably produces different outputs on each press, mimicking the randomness of a real dice. The circuit performed accurately in tests and consistently delivered expected results.

8. Advantages / Applications

- > **Fair and Accurate Results** – The dice generates truly random numbers, avoiding human bias or manipulation.
- > **Compact and Portable** – The circuit is small and can be easily integrated into electronic board games.
- > **Interactive Design** – Includes LEDs, a buzzer, and a display for a fun and engaging user experience.
- > **Educational Value** – Useful for demonstrating concepts in electronics, probability, and programming. Helps in learning microcontroller basics.
- > **Customizable Output** – Can be modified to display numbers, animations, or sound effects.
- > **Ideal for Projects** – A great choice for engineering mini-projects, exhibitions, or hobby builds.

9. Challenges Faced

- Segment connections not displaying complete digits initially.
- Last two LEDs were not lighting up due to pin conflicts.
- Debugging logic for button press and buzzer timing.

10. Conclusion

- > **Simple Yet Effective** – The Electronic Dice project demonstrates how basic electronics can replicate a real-world object efficiently.
- > **Practical Application** – It can be used in place of traditional dice for games, ensuring fairness and reducing physical wear.
- > **Educational Project** – Helps in understanding Arduino programming, circuit design, and digital electronics.
- > **Cost-Effective Solution** – Built using low-cost components, making it accessible for students and hobbyists.
- > **Reliable Output** – Delivers consistent and accurate random numbers at the press of a button.
- > **Encourages Innovation** – Acts as a foundation for more advanced gaming or automation projects.

11. References

- > **YouTube** – such as GreatScott!, Electronics Hub, and Arduino Project Hub
Practical demonstrations and project walkthroughs related to electronic dice and Arduino.

> **ChatGPT (OpenAI)**

Assistance in coding and troubleshooting during project development.